SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	SLOT-D	**Date:**	

Code of Conduct

*I **K.Keerthana Sai(192425143)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: K.Keerthana Sai

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

Field	Soil Condition	Irrigation Status	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- $Dry(x) \land IrrigationAvailable(x) \rightarrow NeedsWater(x)$
- $NeedsWater(x) \rightarrow Irrigate(x)$
- $Wet(x) \rightarrow \neg NeedsWater(x)$
- $IrrigationUnavailable(x) \rightarrow \neg Irrigate(x)$
- $Rice(x) \land NeedsWater(x) \rightarrow PriorityIrrigation(x)$

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

ANSWERS:

1. Frame Semantics in Banking Customer-Service Systems

Introduction

Frame Semantics is an NLP technique used to understand the meaning of a sentence by identifying an action and the participants involved in that action. In banking customer-service systems, semantic frames help the system correctly understand customer requests and perform the required banking operations.

1. Analyze the Semantic Frame of Each Banking Request

Query 1: “Transfer ₹20,000 from my savings account to my son's account.”

Frame: TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)

- Action: Transfer
- Source Account: Savings account
- Amount: ₹20,000

- Destination Account: Son's account
- Customer: User

The customer wants money to be transferred from one account to another.

Query 2: “Block my debit card immediately.”

Frame: BLOCK(DebitCard, Customer)

- Action: Block
- Object: Debit card
- Customer: User

The customer wants the debit card to be blocked immediately.

Query 3: “Send my transaction statement to my email.”

Frame: SEND(TransactionStatement, Email, Customer)

- Action: Send
- Object: Transaction statement
- Destination: Email
- Customer: User

The customer wants the transaction statement to be sent to their email address.

Query 4: “Increase my daily withdrawal limit.”

Frame: MODIFY(WithdrawalLimit, Customer)

- Action: Modify
- Object: Daily withdrawal limit
- Modification: Increase
- Customer: User

The customer wants the existing withdrawal limit to be increased.

2. Identify the Incorrect Semantic Interpretation

The incorrect interpretation is **B4**.

Query ID	Actual User Intent	System Interpretation	Result
B1	Transfer money to another account	Transfer money	Correct
B2	Block debit card	Block debit card	Correct
B3	Receive transaction statement by email	Send statement to email	Correct
B4	Increase withdrawal limit	Decrease withdrawal limit	**Incorrect**

B4 is incorrect because **increase** and **decrease** have opposite meanings. The system has identified the general modification operation but has misunderstood the direction of the modification.

3. Effect of Incorrect Semantic Roles

Incorrect semantic-role identification can lead to an incorrect banking operation. For example, if a customer asks to increase the daily withdrawal limit but the system interprets it as a decrease, the customer's withdrawal limit may be reduced.

This can result in:

- Incorrect banking transactions
- Customer inconvenience
- Financial problems
- Customer complaints
- Loss of trust in the banking system

Therefore, semantic roles must be accurately identified before executing important banking operations.

4. Recommended Frame-Based Semantic Analysis Approach

A reliable banking conversational system should:

1. Identify the main action or frame.
2. Identify all participants involved in the action.
3. Extract important entities such as account numbers, amounts and destinations.
4. Identify modifiers such as increase, decrease and immediately.
5. Validate the semantic roles before executing the operation.
6. Ask the customer for confirmation for sensitive transactions.

Conclusion

Frame-based semantic analysis helps banking systems understand customer requests accurately. Correct identification of actions and participants is essential because even a small semantic error can result in an incorrect financial operation. Therefore, frame-based analysis with validation and confirmation can improve the safety and reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

Introduction

First-Order Predicate Calculus (FOPC) is a logical representation technique used to represent facts, objects and relationships. In smart agriculture, predicate logic can be used to represent soil conditions, irrigation availability and crop requirements and to make automated irrigation decisions.

1. Representation of Field Observations Using FOPC

The field observations can be represented as follows.

Field F1:

- Dry(F1)
- IrrigationAvailable(F1)
- Rice(F1)

Field F2:

- Wet(F2)
- IrrigationAvailable(F2)

- Wheat(F2)

Field F3:

- Dry(F3)
- IrrigationUnavailable(F3)
- Maize(F3)

Field F4:

- Wet(F4)
- IrrigationUnavailable(F4)
- Rice(F4)

The given rules can be represented as:

1. ∀x [Dry(x) ∧ IrrigationAvailable(x) → NeedsWater(x)]
2. ∀x [NeedsWater(x) → Irrigate(x)]
3. ∀x [Wet(x) → ¬NeedsWater(x)]
4. ∀x [IrrigationUnavailable(x) → ¬Irrigate(x)]
5. ∀x [Rice(x) ∧ NeedsWater(x) → PriorityIrrigation(x)]

2. Fields That Require Irrigation

Field F1

F1 is dry and irrigation is available.

Dry(F1) ∧ IrrigationAvailable(F1)

Therefore:

NeedsWater(F1)

Using the second rule:

Irrigate(F1)

Since F1 contains rice:

PriorityIrrigation(F1)

Therefore, F1 should be irrigated with priority.

Field F2

F2 is wet.

Wet(F2)

Therefore:

¬NeedsWater(F2)

Hence, F2 does not require irrigation.

Field F3

F3 is dry, but irrigation is unavailable.

Dry(F3)

and

IrrigationUnavailable(F3)

Therefore:

¬Irrigate(F3)

F3 requires water because the soil is dry, but irrigation cannot be performed because water is unavailable.

Field F4

F4 is wet and irrigation is unavailable.

Wet(F4) → ¬NeedsWater(F4)

Therefore, F4 does not require irrigation.

Irrigation Decision Table

Field	Soil Condition	Irrigation Status	Decision
F1	Dry	Available	**Irrigate**
F2	Wet	Available	**Do not irrigate**
F3	Dry	Unavailable	**Cannot irrigate**
F4	Wet	Unavailable	**Do not irrigate**

3. Should F1 and F3 Be Treated Differently?

Yes, F1 and F3 should be treated differently.

Both fields have dry soil, but their irrigation conditions are different.

F1:

Dry + Irrigation Available → Needs Water → Irrigate

F3:

Dry + Irrigation Unavailable → Cannot Irrigate

Therefore, the controller should irrigate F1, while F3 should be marked as requiring water but currently unavailable for irrigation.

4. Is Predicate Logic Alone Sufficient?

No. Predicate logic is useful when the information is clear and certain. However, real agricultural systems often receive incomplete or uncertain sensor information.

For example, sensors may produce:

- Incorrect readings
- Missing data
- Noisy measurements
- Uncertain moisture levels
- Sensor failures

Basic predicate logic cannot represent uncertainty effectively. Therefore, practical agricultural systems can combine predicate logic with fuzzy logic, probability models, machine learning and sensor-confidence values.

Conclusion

First-Order Predicate Calculus provides a useful method for representing agricultural conditions and making rule-based irrigation decisions. However, real-world agricultural systems require additional techniques to handle incomplete and uncertain sensor information.

3. Word Sense Disambiguation in a Travel Recommendation System

Introduction

Word Sense Disambiguation (WSD) is the process of identifying the correct meaning of an ambiguous word based on its context. Travel recommendation systems use WSD to understand whether a word refers to a travel destination, company, software or another meaning.

1. Determine the Intended Sense of the Ambiguous Terms

Amazon Tour

The user viewed rainforest trekking packages.

Therefore:

Amazon → Amazon Rainforest

Safari Booking

The user selected wildlife resort packages.

Therefore:

Safari → Wildlife Tour

Java Vacation

The user viewed hotels in Bali and Java.

Therefore:

Java → Java Island

Cruise Package

The user viewed Mediterranean ship packages.

Therefore:

Cruise → Ship Journey

Safari Download for Windows

The user clicked a browser download page.

Therefore:

Safari → Web Browser

2. Lexical and Contextual Cues

Query	Important Cues	Intended Sense
Amazon tour	Tour, rainforest, trekking	Amazon Rainforest
Safari booking	Booking, wildlife, resort	Wildlife Tour

Query	Important Cues	Intended Sense
Java vacation	Vacation, hotels, Bali	Java Island
Cruise package	Mediterranean, ship, package	Ship Journey
Safari download for Windows	Download, Windows, browser	Web Browser

The surrounding words provide important information about the intended meaning.

For example, “Safari booking” is related to wildlife tourism, while “Safari download for Windows” is related to a web browser.

3. Why the Same Word Can Have Different Interpretations

The same word can have different meanings for different users because users have different search contexts and interests.

For example:

User1:

Searches “Safari booking” and selects a wildlife resort.

→ Safari means wildlife tour.

User2:

Searches “Safari download for Windows” and clicks a browser download page.

→ Safari means web browser.

Therefore, WSD should consider the user's current query as well as their previous interactions.

4. Industrial-Scale WSD Strategy

A large-scale WSD system can use the following process:

User Query

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Text Preprocessing

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Identify Ambiguous Words

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Generate Possible Senses

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Analyze Query Context

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Analyze User History

↓

Use Word Embeddings

↓

Analyze Click Behaviour

↓

Rank Possible Senses

↓

Generate Personalized Recommendation

The system can combine:

- Query context
- User search history
- User clicks
- Viewed products
- Embeddings
- Previous bookings

Embeddings help the system understand the semantic similarity between the query and different possible meanings. Click behaviour provides additional evidence about what the user actually intended.

Conclusion

Word Sense Disambiguation is important for travel recommendation systems because the same word can have different meanings. By combining query context, user history, embeddings and click behaviour, the system can identify the correct meaning and provide more accurate and personalized recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

Introduction

Syntax-driven semantic analysis uses the syntactic structure of a sentence to determine its meaning and semantic roles. In legal NLP, it is important to correctly identify roles such as Agent, Object, Recipient and Source because incorrect interpretation can affect legal information extraction.

1. Influence of Syntactic Structure on Semantic Roles

Sentence 1

“The supplier delivered the equipment to the buyer.”

This is an active sentence.

- Supplier = Subject
- Delivered = Verb
- Equipment = Object
- Buyer = Recipient

Therefore:

Supplier = Agent

Equipment = Object/Theme

Buyer = Recipient

Sentence 2

"The buyer received the equipment from the supplier."

Here, the buyer is the grammatical subject, but this does not mean that the buyer is the agent.

Correct roles are:

Buyer = Recipient

Equipment = Object/Theme

Supplier = Agent/Source

Sentence 3

"The company terminated the contract after the violation."

This is an active construction.

Company = Agent

Contract = Object/Theme

The company performs the action of terminating the contract.

Sentence 4

"The contract was terminated by the company."

This is a passive construction.

The contract is the grammatical subject, but it does not perform the action.

Correct roles:

Contract = Object/Theme

Company = Agent

2. Incorrect Semantic-Role Assignments

There are two incorrect interpretations in the given data.

Sentence 2

Given interpretation:

- Buyer = Agent
- Supplier = Recipient

This is incorrect.

Correct interpretation:

- Buyer = Recipient
- Supplier = Agent/Source

Sentence 4

Given interpretation:

- Contract = Agent
- Company = Object

This is incorrect.

Correct interpretation:

- Contract = Object/Theme

- Company = Agent

Corrected Semantic Roles

Legal Sentence	Correct Semantic Roles
Supplier delivered equipment to buyer	Supplier = Agent, Equipment = Object, Buyer = Recipient
Buyer received equipment from supplier	Buyer = Recipient, Equipment = Object, Supplier = Agent/Source
Company terminated contract	Company = Agent, Contract = Object
Contract was terminated by company	Contract = Object/Theme, Company = Agent

3. Effect of Confusing Grammatical Subject with Semantic Agent

A grammatical subject is not always the entity performing the action.

For example:

“The contract was terminated by the company.”

The grammatical subject is contract, but the company actually performs the action.

If an NLP system assumes that the subject is always the agent, it may incorrectly conclude:

Contract = Agent

instead of:

Company = Agent

This can lead to incorrect extraction of:

- Responsible parties
- Legal actions
- Contract violations
- Parties involved in agreements
- Recipients and sources of actions

Such errors can be serious in legal applications because the extracted information may be used for contract analysis and legal decision support.

4. Recommended Syntax-Driven Semantic Analysis Architecture

A reliable legal NLP system should use the following pipeline:

Legal Document

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Sentence Segmentation

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Tokenization

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Part-of-Speech Tagging

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Syntactic Parsing

↓

Active/Passive Voice Detection

↓

Predicate Identification

↓

Semantic Role Labeling

↓

Prepositional Phrase Analysis

↓

Coreference Resolution

↓

Long-Distance Dependency Resolution

↓

Legal Entity and Relation Extraction

The system should handle:

1. **Active voice** – identify the entity performing the action.
2. **Passive voice** – identify the agent using the "by" phrase.
3. **Prepositional phrases** – identify roles such as recipient, source and location.
4. **Long-distance dependencies** – connect related words across complex sentences.
5. **Coreference** – resolve words such as "it", "they" and "the party".
6. **Semantic Role Labeling** – assign appropriate semantic roles to the participants.

Conclusion

Syntax-driven semantic analysis is essential for accurately processing legal documents. The grammatical subject cannot always be considered the semantic agent, especially in passive constructions. Combining syntactic parsing, voice detection, semantic role labeling, prepositional phrase analysis and dependency resolution can improve the accuracy and reliability of legal NLP systems.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	4	4	3	3	4	20	90
2	4	4	4	3	3	22	
3	4	4	4	4	4	22	
4	4	3	4	4	3	24	

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